Bhasha Setu - Ultimate Complete Features Guide

Bhasha Setu - Ultimate Complete Features and Implementation Guide

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Live Application URL: https://telugu-hindi-voice-bridge.lovable.app (https://telugu-hindi-voice-bridge.lovable.app/)
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Project Overview

Telugu Hindi Voice Bridge (Bhasha Setu) is a real-time voice and text translation app for seamless communication between Telugu and Hindi speakers. Speak or type in one language and get instant translation, transliteration, and text-to-speech audio.

Complete Features List

1. Side-by-Side Dual Translation Boxes

- Modern side-by-side layout for Telugu and Hindi.
- Inline speaker icons for one-click audio playback.
- Smooth animations and responsive design.

2. Clear Voice Playback

- Crisp, natural-sounding audio feedback.
- Enhanced Web Speech API integration.
- Volume and quality controls.

3. 3D Natural Audio Voices for All Languages

- Distinct male and female high-fidelity voices.
- ElevenLabs or advanced TTS for natural sound.
- Available for every supported language.

4. Enhanced Voice Clarity

- Perfect clarity for both male and female voices.
- No distortion or muffled sound.
- Clear pronunciation whether speaking to male or female listener.

5. Security Features

- Password and email ID required for access.
- No data shared with third parties.

- Encrypted storage and secure authentication.
- Hack protection especially for Android version.

6. Multi-Language Support

- Accurate translation and transliteration.
- Full support for all 22 scheduled Indian languages.

7. Low Data Usage

- Minimal bandwidth consumption.
- Optimized API calls and efficient caching.

8. Cross-Platform Compatibility

- Works on all phones, laptops, and devices.
- PWA structure with responsive design.

9. Offline Support

- Fully usable without internet or data.
- Service worker caching for core features.
- Local processing for translation and voice.

10. Translation History

- Save and view past translations.
- Searchable with timestamps.

11. Dark Mode

- Toggle between light and dark themes.
- Eye-friendly for night use.

12. Auto Language Detection

- Automatically detect input language.
- Seamless switching between Telugu and Hindi.

13. Favorite Phrases

- Save and access commonly used phrases quickly.
- Personalized list for frequent translations.
- One-tap to use in translation.

14. Speech Speed Control

- Adjust playback speed for voice output.
- Slower or faster options for better understanding.

- Slider or preset controls.

15. Copy and Share Option

- Easy one-click copy of translated text.
- Share translations via social media or messaging apps.
- Export options for notes or documents.

16. Pronunciation Practice Mode

- Practice speaking with real-time feedback.
- Compare your pronunciation to correct audio.
- Scoring and improvement tips.

Technical Architecture

- Frontend: React 19, TypeScript
- Design: Tailwind CSS, shadcn/ui, Lucide Icons
- Speech: Web Speech API, ElevenLabs TTS
- Translation: Server-side engine
- Offline & Cross-Platform: PWA with service workers
- Security: Encrypted auth backend
- Data Optimization: Efficient caching
- Additional: Local storage for favorites and history

Implementation Roadmap

1. Brand renaming to Bhasha Setu.
2. Implement PWA and offline caching.
3. Add security layer and low data optimizations.
4. Integrate 3D voices and clarity enhancements.
5. Add history, dark mode, auto detection.
6. Add favorite phrases, speech speed, copy share, practice mode.
7. Expand to all Indian languages.
8. Android app build with hack protection.
9. Testing and deployment.

Ultimate guide with all features compiled from user requirements.